MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M5, METAL PARTS ASSEMBLY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification is applicable to the metal parts assembly for Catapult, Aircraft Ejection Seat, M5. Assembly will ultimately be loaded and used in conjunction with other cartridge actuated devices in air-crew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specication.

SPECIFICATIONS

MILITARY

MIL–G–2550 — General Specification for Ammunition, Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL–STD–8 — Dimensioning and Toleranceing.

MIL–STD–9 — Screw Thread Conventions and Methods of Specifying.

MIL–STD–10 — Surface Roughness, Waviness and Lay.

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–413 — Visual Inspection Guide for Rubber O-Rings.

ORDNANCE CORPS

URAX6 — Ordnance Finishes (Surface Roughness).

30–1–7 — Standard for Dimensioning and Toleranceing.

DRAWINGS

ORDNANCE CORPS

94–1–24 — Catapult, Aircraft Ejection Seat, M5, Metal Parts Assembly.

F8797377 — Ammunition Con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

tainer, Fiber M299 for Catapult, Aircraft Ejection Seat, M5, Assembly, Details and Packing.

F8797378 — Box, Packing, Ammunition for Catapult, Aircraft Ejection Seat, M5, Assembly, Details and Packing.

GL8655334 — Inspection Equipment List for Catapult, Aircraft Ejection Seat, M5, Metal Parts.

PUBLICATIONS

NATIONAL BUREAU OF STANDARDS

Handbook H28— Screw-Thread Standards for Federal Services.

(Copies of specifications, standards drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Material, parts and assemblies shall meet the requirements specified on the applicable drawings, referenced specifications and the following.

3.2 Preproduction sample. Unless otherwise directed by the contracting officer, a preproduction sample as detailed in 4.2 shall be required.

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Forgings. Forgings shall comply with the requirements of the applicable specification except that the hardness of the unmachined forging shall be as determined by the contractor.

3.5 Physical properties. Physical properties required by drawing shall be uniform throughout each part. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.6 Hydrostatic pressure. The outside, telescoping and inside tubes shall withstand a constant internal pressure as indicated in table 1 for a period of 15 seconds, minimum, without leakage or any mechanical failure (see 6.1).

TABLE I

Component	Hydrostatic pressure pounds per square inch (PSI) minimum
Outside tube	3400
Telescoping tube	2500
Inside tube	3100

3.7 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 400 pounds ± 50 pounds is applied at the steady rate of .25 inches per minute.

3.8 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled "hand tight".

3.9 Structural strength. The metal parts assembly shall be capable of withstanding each test load for 15 seconds, minimum, as indicated in table II. The maximum change between mounting points, resulting from each load application shall not exceed the values indicated in table II. A preload of 500 pounds shall be utilized prior to each test load application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

TABLE II

Test loads minimum	Change between mounting points maximum
1. 2000 pounds in tension	.005 inches
2. 2000 pounds in tension	.001 inches
3. 8000 pounds in compression ..	.005 inches
4. 8000 pounds in compression ..	.001 inches

3.10 Protective finish. Where protective finish is required by components details, finish requirements shall apply to all surfaces of the component regardless of contour configuration.

3.11 Residual magnetism. The metal parts assembly shall not deflect an approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.12 Workmanship. The requirements for workmanship are as specified on the applicable drawings, specifications and the following:

3.12.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.12.2 *Surface finish.* The requirements for surface finish are as detailed by the applicable drawing and Standard MIL–STD–10. The Ordnance Standard Finish Blocks prescribed in URAX6 shall be used as a basis of comparison for surface roughness determination.

3.12.3 *Threads.* Threads shall be full and undamaged for the entire minimum length of depth as required by the applicable drawing and Specification MIL–G–2550.

3.12.4 *Burrs.* No part shall contain a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.12.5 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions contained in this section are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL–STD–109 to define inspection terms used. Inspection for all thread characteristics shall be in accordance with Standard MIL–STD–9 and/or Handbook H28. The Ordnance Corps standard for dimensioning and tolerancing, 30–1–7, shall be used in conjunction with Standard MIL–STD–8 for drawing interpretation. Inspection requirements shall be in accordance with Specification MIL–G–2550 except as specified herein. Inspection for "O" rings shall be in accordance with Standard MIL–STD–413.

4.1.1 *Contractor inspection.* Unless otherwise specified, the inspection required by this specification shall be performed by the contractor.

4.1.2 *Contractor Quality Assurance System.* The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under this specification. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification and other referenced documents. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification or other referenced documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.1.3 *Government verification.* All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of monitoring the operations to determine whether the practices, methods and procedures of the written inspection plan, as provided in the quality assurance system, are being properly applied. Verification may also consist of Government product inspection to measure the quality of the product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to the prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction inspection. Prior to the commencement of regular production, the contractor shall submit a preproduction sample to a designated pilot lot laboratory for approval. The sample shall consist of 5 assemblies, 10 sets of unassembled parts and 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the preproduction sample shall be inspected at the pilot lot laboratory for all requirements of the drawings and specifications including all those characteristics listed in 4.3.3 and the tests in 4.3.5. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions. Alternative inspection procedures and inspection equipment may be used by the contractor when such procedures and equipment provide, as a minimum, quality assurance required in the contractual documents. The inspection procedures or equipment specified in this specification will prevail in case of dispute.

4.3.1 *Material.* Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification of the material and part, the applicable requirements, the specific test performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction sample. Prior to inspection, the inspector shall determine from the information stated on the test reports whether the material submitted complies with requirements. Components made from material which does not comply shall be rejected.

4.3.2 *Lot.*

4.3.2.1 *Submittal of product.* The contractor shall submit the product in accordance with Standard MIL–STD–105.

4.3.2.2 *Size of lot.* For the purposes of identification and shipment the first shipping lot of metal parts assemblies shall consist of 615 catapults, maximum, and all subsequent lots shall consist of 610 catapults, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.3.3 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in tables III and IV. For an individual defect (single characteristic) the criteria in table III shall be used. For total defects within a major or minor group the criteria in table IV shall be used. All nonconforming material shall be rejected.

TABLE III

Lot size	Sample size	Accept numbers	
		Major	Minor
1 — 300	100% examination		
301 — 1300	110	1	2
1301 — 3200	150	2	3
3201 — 8000	225	3	4
8001 — 22,000	300	4	5

TABLE IV

No. of defects in class	Total allowable defects per sample size			
	Sample sizes for major defects			
	110	150	225	300
	Acceptance numbers			
2 — 3	1	2	3	4
4 — 7	2	3	4	5
8 — 19	3	4	5	7
20 — 43	4	5	8	10
over 43	6	8	11	14
No. of defects in class	Sample sizes for minor defects			
	110	150	225	300
	Acceptance numbers			
2 — 3	2	3	4	5
4 — 7	3	4	5	7
8 — 16	4	5	8	10
17 — 48	6	8	11	14
over 48	8	11	17	20

4.3.3.1 *Seal, fixed, Pcmk 94–0–1C.*

Defect	*Method of inspection*
Major	
1. Diameter of wire, max	Measure.
2. Outside diameter, max	Measure.
3. Length of tang, incorrect	Measure.
4. Locational variance of ends, max	Visual.
5. Number of coils, incorrect	Visual.
6. Metal defective	Visual.
7. Protective finish inadequate or defective	Visual.
8. Burr or sharp edge	Visual.
9. Foreign matter	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

Minor

41.	Outside diameter, min	Measure.
42.	Length between coils, incorrect	Measure.
43.	Radius of tang bend, incorrect	Measure.
44.	Direction of turn, incorrect	Visual.
45.	Radius and/or chamfer missing or incorrect	Visual.

4.3.3.2 *Seal, plain, Pcmk 94-0-1E.*

	Defect	*Method of inspection*
Major		
1.	Diameter of wire, max	Measure.
2.	Outside diameter, min	Measure.
3.	Locational variance of ends, max	Visual.
4.	Number of coils, incorrect	Visual.
5.	Metal defective	Visual.
6.	Protective finish inadequate or defective	Visual.
7.	Burr or sharp edge	Visual.
8.	Foreign matter	Visual.
Minor		
41.	Outside diameter, max	Measure.
42.	Length between coils, incorrect	Measure.
43.	Direction of turn, incorrect	Visual.
44.	Radius and/or chamfer missing or incorrect	Visual.

4.3.3.3 *Ring, stop, Pcmk 94-1-10A.*

	Defect	*Method of inspection*
Critical		
a.	Inside diameter, min	Gage.
Major		
1.	Pitch diameter of thread, min	Gage.
2.	Major diameter of thread, min	Gage.
3.	Concentricity of inside diameter with pitch diameter of thread	Gage.
4.	Concentricity of counterbore with inside diameter	Gage.
5.	Diameter of thread undercut, min	Gage.
6.	Length to flange	Gage.
7.	Length to counterbore	Gage.
8.	Length to datum diameter	Gage.
9.	Angle at datum diameter	Gage.
10.	Thread damaged or not full	Visual.
11.	Metal defective	Visual.
12.	Protective finish inadequate or defective	Visual.
13.	Surface finish of inside diameter or chamfer on inside diameter, improper	Visual.
14.	Burr or sharp edge	Visual.
15.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of thread, max	Gage.
42.	Inside diameter, max	Gage.
43.	Diameter of counterbore	Gage.
44.	Diameter of flange hole	Gage.
45.	Total length	Gage.
46.	Diameter of flange	Measure.
47.	Length between flange holes	Measure.
48.	Location of flange holes	Measure.
49.	Depth of flange hole	Visual.
50.	Length of thread undercut, max	Visual.
51.	Radius and/or chamfer missing or incorrect	Visual.
52.	Surface finish improper except as otherwise classified	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.3.3.4 *Latch, Pcmk 94-1-10B.*

Defect	Method of inspection
Critical	
a. Metal defective	Visual.
Major	
1. Diameter of firing pin relief	Gage.
2. Total length	Gage.
3. Length to inside surface at small end	Gage.
4. Location of firing pin relief	Gage.
5. Length of cutout	Gage.
6. Width at small end	Gage.
7. Width at cutout	Gage.
8. Thickness at datum location	Gage.
9. Angle at large end	Gage.
10. Chamfer on inside surface at small end missing or incorrect	Measure.
11. Protective finish inadequate or defective	Visual.
12. Surface finish improper	Visual.
13. Burr or sharp edges	Visual.
14. Foreign matter	Visual.
Minor	
41. Thickness	Gage.
42. Thickness at firing pin relief	Gage.
43. Width at large end	Gage.
44. Radial profile or large end	Gage.
45. Radial profile of sides	Measure.
46. Raidus and/or chamfer missing or incorrect except as otherwise classified	Visual.

4.3.3.5 *Block, Pcmk 94-1-11A.*

Defect	Method of inspection
Critical	
a. Pipe thread hole obstructed	Visual.
Major	
1. Pitch diameter of thread, max	Gage.
2. Minor diameter of thread, max	Gage.
3. Pitch diameter of pipe thread, max	Gage.
4. Concentricity of counterbore with small cavity	Gage.
5. Concentricity of pitch diameter of thread with counterbore	Gage.
6. Concentricity of large cavity with counterbore	Gage.
7. Concentricity of pitch diameter of thread with body	Gage.
8. Concentricity of body shoulder with body	Gage.
9. Concentricity of flange with body shoulder	Gage.
10. Symmetry of pin hole with small cavity	Gage.
11. Symmetry of width of slot with body shoulder	Gage.
12. Symmetry of thickness of rectangular section with body shoulder	Gage.
13. Symmetry of hole in rectangular section with body shoulder	Gage.
14. Symmetry of width of rectangular section with body shoulder	Gage.
15. Diameter of small cavity	Gage.
16. Diameter of cavity counterbore, max	Gage.
17. Diameter of hole in rectangular section	Gage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

	Defect	Method of inspection
Major		
18.	Depth of large cavity	Gage.
19.	Length of effective thread, min	Gage.
20.	Length from top of flange to face of base	Gage.
21.	Length of slot in body	Gage.
22.	Length to flange, max	Gage.
23.	Width of rectangular section	Gage.
24.	Thickness of rectangular section	Gage.
25.	Location of body slot to base of large cavity	Gage.
26.	Diameter of thread undercut, max	Measure.
27.	Profile of end of rectangular section	Measure.
28.	Threads damaged or not full	Visual.
29.	Metal defective	Visual.
30.	Protective finish inadequate or defective	Visual.
31.	Surface finish of small cavity, body shoulder or body slots improper	Visual.
32.	Burr	Visual.
33.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of thread, min	Gage.
42.	Pitch diameter of pipe thread, min	Gage.
43.	Diameter of large cavity counterbore, min	Gage.
44.	Diameter of large cavity	Gage.
45.	Diameter of body	Gage.
46.	Diameter of body shoulder	Gage.
47.	Diameter of flange	Gage.
48.	Diameter of pin hole in body	Gage.
49.	Diameter of drilled hole at bottom of small cavity	Gage.
50.	Diameter of holes at open end	Gage.
51.	Depth to bottom of small cavity	Gage.
52.	Depth to bottom of drilled hole	Gage.
53.	Length to flange, min	Gage.
54.	Length over flange	Gage.
55.	Length to hole in rectangular section	Gage.
56.	Total length	Gage.
57.	Width of slot in body	Gage.
58.	Location of pin hole in body	Gage.
59.	Diameter of undercut on body shoulder, min	Measure.
60.	Length of chamfer at small cavity	Measure.
61.	Location of hole through open end	Visual.
62.	Location of pipe hole thread	Visual.
63.	Depth of cavity counterbore, min	Visual.
64.	Radius and/or chamfer missing or incorrect	Visual.
65.	Surface finish improper except as otherwise classified	Visual.

4.3.3.6 *Washer, shock, Pcmk 94-1-12D.*

	Defect	Method of inspection
Major		
1.	Tear, crack or split	Visual.
Minor		
41.	Thickness	Measure.
42.	Flash	Visual.
43.	Foreign matter	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.3.3.7 *Pin, shear, Pcmk 94-1-13A.*

	Defect	Method of inspection
Major		
1.	Diameter, max	Gage.
2.	Total length, min	Gage.
3.	Metal defective	Visual.
4.	Protective finish inadequate or defective	Visual.
5.	Surface finish improper	Visual.
6.	Burr	Visual.
7.	Foreign matter	Visual.
Minor		
41.	Total length, max	Gage.
42.	Profile of end, incorrect	Visual.

4.3.3.8 *Plug, Pcmk 94-1-13B.*

	Defect	Method of inspection
Critical		
a.	Diameter of boss hole, min	Gage.
Major		
1.	Concentricity of bass hole with cavity	Gage.
2.	Concentricity of flange with cavity	Gage.
3.	Concentricity of body with flange	Gage.
4.	Symmetry of pin hole with cavity	Gage.
5.	Symmetry of base slot with cavity	Gage.
6.	Diameter of cavity	Gage.
7.	Diameter of flange, min	Gage.
8.	Depth of cavity, min	Gage.
9.	Depth of base slot	Gage.
10.	Total length, max	Gage.
11.	Chamfer on base of cavity missing or incorrect	Visual.
12.	Metal defective	Visual.
13.	Protective finish inadequate or defective	Visual.
14.	Surface finish of cavity or surfaces of base slot improper	Visual.
15.	Burr	Visual.
16.	Foreign matter	Visual.
Minor		
41.	Diameter of boss hole, max	Gage.
42.	Diameter of flange, max	Gage.
43.	Diameter of body	Gage.
44.	Diameter of pin hole in body	Gage.
45.	Diameter of undercut, min	Gage.
46.	Depth of cavity, max	Gage.
47.	Total length, min	Gage.
48.	Length from slotted end to pin hole in body	Gage.
49.	Length from slotted end to flange	Gage.
50.	Width of base slot	Gage.
51.	Height of boss	Gage.
52.	Diameter of boss	Measure.
53.	Angle of boss	Measure.
54.	Radius and/or chamfer missing or incorrect except as otherwise classified	Visual.
55.	Surface finish improper except as otherwise classified	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.3.3.9 *Pin, firing, Pcmk 94-1-13C.*

	Defect	Method of inspection
Critical		
a.	Diameter of striker, max	Gage.
Major		
1.	Concentricity of striker and groove flange with striker shoulder	Gage.
2.	Concentricity of groove with groove flange	Gage.
3.	Symmetry of width at forward datum location and width of body with striker shoulder	Gage.
4.	Symmetry of thickness of body and hole in striker shoulder with striker shoulder and location of hole in striker shoulder	Gage.
5.	Diameter of striker shoulder, min	Gage.
6.	Diameter of groove flange, min	Gage.
7.	Diameter of groove	Gage.
8.	Depth of hole in striker shoulder, max	Gage.
9.	Length of striker	Gage.
10.	Length across groove flange	Gage.
11.	Width at forward datum location	Gage.
12.	Width at rear datum location	Gage.
13.	Width of body	Gage.
14.	Thickness of body	Gage.
15.	Relief grooves on striker shoulder missing or incorrect	Visual.
16.	Metal defective	Visual.
17.	Protective finish inadequate or defective	Visual.
18.	Surface finish, improper	Visual.
19.	Burr	Visual.
20.	Foreign matter	Visual.
Minor		
41.	Diameter of striker, min	Gage.
42.	Diameter of striker shoulder, max	Gage.
43.	Diameter of groove flange, max	Gage.
44.	Diameter of hole in striker shoulder	Gage.
45.	Depth of hole in striker shoulder, min	Gage.
46.	Total length	Gage.
47.	Length from grooved end to striker shoulder	Gage.
48.	Length across relief grooves of striker shoulder, min.	Gage.
49.	Surface variation on rear face of striker shoulder	Gage.
50.	Diameter of lathe center hole, max	Measure.
51.	Length of rear flange of groove	Measure.
52.	Length of groove	Measure.
53.	Depth of relief groove or striker shoulder, min	Visual.
54.	Radius and/or chamfer missing or incorrect	Visual.

4.3.3.10 *Tube, outside, Pcmk 94-1-26A.*

	Defect	Method of inspection
Critical		
a.	Metal defective	Visual.
Major		
1.	Pitch diameter of thread, min	Gage.
2.	Major diameter of thread, min	Gage.
3.	Concentricity of pitch diameter of thread with inside diameter at datum location	Gage.
4.	Outside diameter, min	Gage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

	Defect	Method of inspection
Major		
5.	Inside diameter, max	Gage.
6.	Total length, min	Gage.
7.	Thickness of wall, min	Gage.
8.	Straightness	Gage.
9.	Length of effective thread, min	Measure.
10.	Thread damaged or not full	Visual.
11.	Protective finish inadequate or defective	Visual.
12.	Surface finish improper	Visual.
13.	Burr	Visual.
14.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of thread, max	Gage.
42.	Outside diameter, max	Gage.
43.	Inside diameter, min	Gage.
44.	Total length, max	Gage.
45.	Length to end of notch	Measure.
46.	Length of thread, max	Measure.
47.	Depth of notch	Visual.
48.	Width of notch	Visual.
49.	Chamfer of thread missing or incorrect	Visual.

4.3.3.11 *Tube, telescoping, Pcmk 94-1-26B.*

	Defect	Method of inspection
Critical		
a.	Outside diameter of body, max	Gage.
b.	All holes in shoulder missing	Visual.
c.	Metal defective	Visual.
Major		
1.	Outside diameter of body, min	Gage.
2.	Diameter of shoulder, min	Gage.
3.	Inside diameter, max	Gage.
4.	Total length, max	Gage.
5.	Variation of wall thickness, max	Gage.
6.	Straightness	Gage.
7.	Number of holes in shoulder incorrect	Visual.
8.	Protective finish inadequate or defective	Visual.
9.	Surface finish improper	Visual.
10.	Burr	Visual.
11.	Foreign matter	Visual.
Minor		
41.	Diameter of shoulder, max	Gage.
42.	Inside diameter, min	Gage.
43.	Total length, min	Gage.
44.	Length of shoulder	Gage.
45.	Diameter of holes in shoulder	Measure.
46.	Location of holes in shoulder	Visual.
47.	Spacing of holes in shoulder incorrect	Visual.

4.3.3.12 *Tube, inside, Pcmk 94-1-27A.*

	Defect	Method of inspection
Critical		
a.	Metal defective	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

	Defect	Method of inspection
Major		
1.	Pitch diameter of thread, min	Gage.
2.	Major diameter of thread, min	Gage.
3.	Concentricity of pitch diameter of thread with outside diameter at datum location	Gage.
4.	Concentricity of pitch diameter of thread with inside diameter at datum location	Gage.
5.	Outside diameter of body at ends	Gage.
6.	Inside diameter, max	Gage.
7.	Depth of spiral groove, max	Gage.
8.	Depth of end relief of spiral groove	Gage.
9.	Length of effective thread, min	Gage.
10.	Total length	Gage.
11.	Wall thickness, min	Gage.
12.	Straightness	Gage.
13.	Threads damaged or not full	Visual.
14.	Protective finish inadequate or defective	Visual.
15.	Surface finish of outside diameter of body at ends	Visual.
16.	Burr	Visual.
17.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of thread, max	Gage.
42.	Inside diameter, min	Gage.
43.	Outside diameter at midsection, max	Gage.
44.	Depth of spiral groove, min	Gage.
45.	Depth of slot	Gage.
46.	Length of shoulder	Measure.
47.	Length of slot, max	Measure.
48.	Length of flat on slot, min	Measure.
49.	Length of thread, max	Measure.
50.	Width of slot	Measure.
51.	Location of spiral groove	Measure.
52.	Profile of spiral groove	Visual.
53.	Direction of turn of spiral groove	Visual.
54.	Number of turns of spiral groove incorrect	Visual.
55.	Spacing of slots incorrect	Visual.
56.	Radius and/or chamfer missing or incorrect	Visual.
57.	Surface finish improper except as otherwise classified	Visual.

4.3.3.13 *Cap, Pcmk 94–1–27B.*

	Defect	Method of inspection
Major		
1.	Pitch diameter of thread, max	Gage.
2.	Minor diameter of thread, max	Gage.
3.	Concentricity of outside diameter with pitch diameter of thread	Gage.
4.	Diameter of thread undercut, max	Gage.
5.	Outside diameter, min	Gage.
6.	Thickness of base	Gage.
7.	Depth of base hole, max	Measure.
8.	Length of thread undercut, max	Measure.
9.	Thread damaged or not full	Visual.
10.	Metal defective	Visual.
11.	Protective finish inadequate or defective	Visual.
12.	Burr	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

	Defect	Method of inspection
Major		
13.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of thread, min	Gage.
42.	Outside diameter, max	Gage.
43.	Diameter of base hole	Measure.
44.	Depth of base hole, min	Measure.
45.	Length between base holes	Measure.
46.	Total length	Measure.
47.	Location of base hole	Measure.
48.	Radius and/or chamfer missing or incorrect	Visual.
49.	Surface finish improper	Visual.

4.3.3.14 *Trunnion, Pcmk 94–1–28A.*

	Defect	Method of inspection
Major		
1.	Pitch diameter of small thread, max	Gage.
2.	Minor diameter of small thread, max	Gage.
3.	Pitch diameter of large thread, max	Gage.
4.	Minor diameter of large thread, max	Gage.
5.	Concentricity of pitch diameter of large thread with inside diameter	Gage.
6.	Concentricity of inside diameter with pitch diameter of small thread	Gage.
7.	Symmetry of lugs	Gage.
8.	Symmetry of width of flats across root of lugs with inside diameter	Gage.
9.	Symmetry of lug with inside diameter	Gage.
10.	Inside diameter	Gage.
11.	Diameter of lug	Gage.
12.	Diameter of small thread undercut, max	Gage.
13.	Diameter of large thread undercut, max	Gage.
14.	Depth of small thread cavity, min	Gage.
15.	Depth of large thread cavity, min	Gage.
16.	Length of effective thread, min	Gage.
17.	Width across flats at root of lugs	Gage.
18.	Symmetry of tang hole with spiral groove	Visual.
19.	Thread damaged or not full	Visual.
20.	Metal defective	Visual.
21.	Protective finish inadequate or defective	Visual.
22.	Surface finish of inside diameter improper	Visual.
23.	Burr	Visual.
24.	Foreign matter	Visual.
Minor		
41.	Pitch diameter of small thread, min	Gage.
42.	Pitch diameter of large thread, min	Gage.
43.	Diameter of small end	Gage.
44.	Depth of small thread cavity, max	Gage.
45.	Depth of large thread cavity, max	Gage.
46.	Depth of spiral groove	Gage.
47.	Total length	Gage.
48.	Length to lug	Gage.
49.	Diameter at large end	Measure.
50.	Diameter of pin hole at spiral groove	Measure.
51.	Length of large diameter	Measure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

	Defect	Method of inspection
Minor		
52.	Location of spiral groove	Measure.
53.	Height of lug	Measure.
54.	Length of large thread undercut, max	Visual.
55.	Profile of spiral groove incorrect	Visual.
56.	Direction of spiral groove incorrect	Visual.
57.	Number of turns of spiral groove incorrect	Visual.
58.	End relief of spiral groove incorrect	Visual.
59.	Notch at small end incorrect	Visual.
60.	Radius and/or chamfer missing or incorrect	Visual.
61.	Surface finish improper except as otherwise classified	Visual.

4.3.3.15 *Metal parts assembly, Pcmk 94–1–24A.*

	Defect	Method of inspection
Critical		
a.	Any part missing	Visual.
b.	Marking incorrect, incomplete or illegible	Visual.
Major		
1.	Block cannot be rotated counterclockwise	Manual.
2.	Cap or stop ring cannot be fully seated	Visual/Manual.
3.	Improper assembly	Visual.
4.	Thread damaged	Visual.
5.	Burr	Visual.
6.	Foreign matter	Visual.
Minor		
41.	Shipping plug missing	Visual.
42.	Coating over marking inadequate or defective	Visual.
43.	Protective finish inadequate or defective	Visual.

4.3.3.16 *Packing in fiber container, Dwg.F8797377.*

	Defect	Method of inspection
Major		
	None.	
Minor		
41.	Gap between body and cover, max	Manual.
42.	Contents loose	Manual.
43.	Metal ends loose or distorted	Visual/Manual.
44.	Cut or damaged through all impregnated layers	Visual.

4.3.3.17 *Packing in wood box, Dwg. F8797378.*

	Defect	Method of inspection
Major		
	None.	
Minor		
41.	Contents loose	Manual.
42.	Hardware missing, loose, broken or improperly engaged	Visual/Manual.
43.	Strapping missing, loose broken or improperly assembled	Visual/Manual.
44.	Contents of box exposed	Visual.
45.	Box damaged	Visual.
46.	Marking misleading or unidentifiable	Visual.
47.	Number of units incorrect	Visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

4.3.4 *Visual examination.* Four power magnification may be used in performing visual examination for such defects as burrs and foreign matter.

4.3.5 *Inspection testing.* The follwing tests shall be performed in accordance with the provisions of 4.4 except as provided in 4.3.

4.3.5.1 *Forgings.*

4.3.5.1.1 *Chemical and physical.* One forging from each heat treated lot of forgings shall be selected for chemical analysis and physical properties. Failure of the test to comply with the acceptance criteria of the applicable specification shall be cause for rejection of all units produced from that heat of forgings.

4.3.5.1.2 *Quality.* Each forging shall be inspected for quality in accordance with the method of the applicable specification. Failure of any forging to comply with the acceptance criteria shall be cause for rejection of the forging.

4.3.5.2 *Hardness.* Fifty units from each heat treated lot of components shall be tested for hardness. Failure of any unit to comply with the requirement on the applicable drawing shall be cause for rejection of the lot of components.

4.3.5.3 *Hydrostatic pressure.* Each outside, telescoping and inside tube shall be subjected to the hydrostatic pressure test. Failure of any tube to comply with the requirements of 3.6 shall be cause for rejection of the tube.

4.3.5.4 *Shear pin strength.* Fifty shear pins from each lot shall be subjected to the shear pin test. Failure of any shear pin to meet the requirements of 3.7 shall be cause for rejection of the lot.

4.3.5.5 *Structural strength.* A sample of 15 catapults shall be selected from each lot and tested for compliance with the requirements of 3.9. Each unit of the sample shall be placed in a tensile testing machine or other approved testing device and subjected to the specified tension and compression loads. The test shall be performed as follows:

(a) Apply a preload of 500 pounds in tension and adjust the dial indicator to zero inches.

(b) Apply the load in tension so as to move the mounting points away from each other.

(c) Release the load in tension to 500 pounds in tension and record the dial indicator reading.

Repeat steps "b" and "c"

(d) Apply a preload of 500 pounds in compression and adjust the dial indicator to zero inches.

(e) Apply the load in compression so as to move the mounting points toward each other.

(f) Release the compression load to 500 pounds in compression and record the dial indicator reading.

Repeat (e) and (f) Above

Should the first reading at the 500 pound preload in either tension or compression exceed .005 inches on any sample or should the second 500 pound preload reading in tension or compression exceed the first reading by more than .001 as permitted in 3.9 the lot shall be rejected.

4.3.5.6 *Protective finish.* The quantity to be tested shall be as determined by the applicable specification. Failure of the sample to comply with the acceptance criteria of the applicable specification shall be cause for rejection of the lot.

4.3.5.7 *Residual magnetism.* Each assembly shall be tested for excessive residual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

magnetism. Failure of any unit to meet the requirement of 3.11 shall be cause for rejection of the unit.

4.3.6 *Inspection equipment.* The examination and tests shall be made using equipment listed on GL8655334, except as provided in 4.3. The Government will determine by means of its monitoring program, that the contractor has available and correctly utilizes, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards, and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test methods and procedures.

4.4.1 *Forgings.*

4.4.1.1 *Chemical and physical.* The forging shall be subjected to the test for chemical analysis and physical properties in accordance with the applicable specification.

4.4.1.2 *Quality.* Prior to the application of protective finish and in the finished condition, the forging shall be inspected for quality in accordance with the method of the applicable specification.

4.4.2 *Hardness.* The sample shall be tested for the hardness values specified on the applicable drawings, using a Rockwell Testing Machine or other approved device.

4.4.3 *Hydrostatic pressure.* The tube shall be mounted in an approved device and subjected to the test for requirements provided in 3.6. The mounting device shall internally seal the ends of the tube within 1½ inches of each end without the application of any axial load other than that applied by the liquid pressure.

4.4.4 *Shear pin strength.* After application of protective finish the shear pin shall be subjected to the shear pin strength test in accordance with 3.7, using the specified fixture in conjunction with a tensile testing machine or other approved device

4.4.5 *Structural strength.* The catapult shall be placed in a tensile testing machine and subjected to the tension and compression loads specified in 3.9. Measurement of the distance between the mounting points shall be at 500 pounds preload prior to and after the required loads have been applied. The load in tension shall be applied on the mounting point so as to move the mounting points away from each other, followed by the compression load so as to move the mounting points toward each other.

4.4.6 *Protective finish.* The test procedure for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

4.4.7 *Residual magnetism.* The catapult shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the catapult 6 inches from and at the same height as an approved compass, in a north-south horizontal position (trunnion end at south) with the compass set in an east-west heading as shown in figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking. Pack-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.

ing, labeling and marking shall be in accordance with Drawings F8797377 and F87-97378.

5.2 **Data cards.** Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

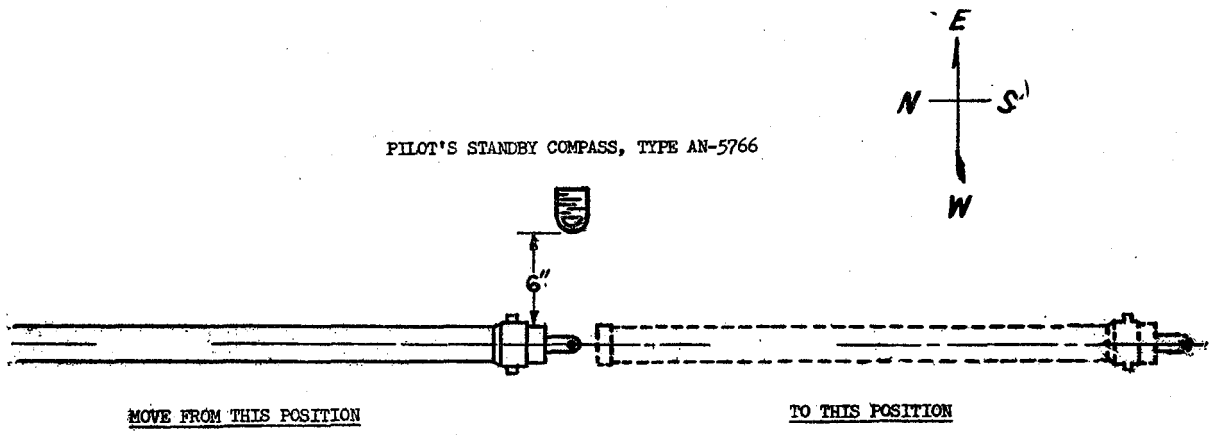
6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Notice. When Government drawings, specifications, or other data, are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded, by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Air Force

Preparing activity:
Army—Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.



NOTE:-

1. CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. *Layout for determination of residual magnetism in catapult, M5.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:32Z
Check the source to verify that this is the current version before use.